

Домашнее задание 15 Гареевой Фариды

Профилирование запросов

Цель:

изучить использование индексов при запросах;
научиться анализировать текстовые логи запросов;
овладеть инструментами профилирования запросов в ClickHouse;

Описание/Пошаговая инструкция выполнения домашнего задания:

1. Возьмите любой демонстрационный DATASET: <https://clickhouse.com/docs/en/getting-started/example-datasets>.
2. Выполните два запроса:
Один с условием WHERE, не использующим первичный ключ (например, с фильтрацией по другому столбцу).
Другой с условием WHERE, использующим первичный ключ (например, с фильтрацией по столбцу, являющемуся РК).
3. Сравните текстовые логи запросов. Выделите строки, относящиеся к пробегу основного индекса в логах запросов.
4. Используйте команду EXPLAIN для анализа выполнения запросов, покажите использование индекса в выводе.

На проверку отправьте текстовый документ с запросами, текстовыми логами и выводом EXPLAIN.

```
create database hw15;

CREATE TABLE hw15.transactions (
    transaction_id UInt32,
    user_id UInt32,
    product_id UInt32,
    quantity UInt8,
    price Float32,
    transaction_date Date
) ENGINE = MergeTree()
ORDER BY (user_id, product_id);

insert into
hw15.transactions(transaction_id,user_id,product_id,quantity,price,transaction_date
)
select number as transaction_id,
rand() % 10000 as user_id,
rand() % 500 as product_id,
10+(rand() % 300) as quantity,
(10000+(rand() % 100000))/100 as price,
toDate(toInt32(today())-(rand() % 365)) as transaction_date

from numbers(1,10000000);
```

Запрос к полю из PK

explain indexes=1

select * from hw15.transactions where user_id=777;

Запрос к полю, по которому нет индекса

explain indexes=1

select * from hw15.transactions where transaction_date='2025-03-08';

Запрос к полю, по которому нет индекса:

```
clickhouse-server-hw13 :) EXPLAIN indexes=1 select * from hw15.transactions where transaction_date='2025-03-08';

EXPLAIN indexes = 1
SELECT *
FROM hw15.transactions
WHERE transaction_date = '2025-03-08'

Query id: bd8ff4bc-6af1-424c-a227-4c4ed4847a62

explain
1. Expression ((Project names + Projection))
2. Expression
3. ReadFromMergeTree (hw15.transactions)
4. Indexes:
5. PrimaryKey
6. Condition: true
7. Parts: 9/9
8. Granules: 123296/123296

8 rows in set. Elapsed: 0.014 sec.
```

Запрос к полю, по которому есть индекс (PK):

```
clickhouse-server-hw13 :) EXPLAIN indexes=1 select * from hw15.transactions where user_id=777;

EXPLAIN indexes = 1
SELECT *
FROM hw15.transactions
WHERE user_id = 777

Query id: 25eacda4-f1ac-40a0-8ce9-18b46bde4fab

Connecting to localhost:9000 as user default.
Connected to ClickHouse server version 25.4.4.

explain
1. Expression ((Project names + Projection))
2. Expression
3. ReadFromMergeTree (hw15.transactions)
4. Indexes:
5. PrimaryKey
6. Keys: _
7. user_id
8. Condition: (user_id in [777, 777])
9. Parts: 9/9
10. Granules: 11/123296
```

Установим уровень логирования trace

SET send_logs_level='trace';

Вывод логов по запросу, в котором используется поиск по индексу PK (всего 67 строк)

```

2 Debug> executeQuery: (from 127.0.0.1:48116) (query 1, line 1) EXPLAIN indexes=1 select * from hw15.transactions where user_id=777; (stage: Complete)
3 Trace> Planner: Query to stage Complete
4 Trace> Planner: Query from stage FetchColumns to stage Complete
5 Trace> QueryPlanOptimizePrewhere: The min valid primary key position for moving to the tail of PREWHERE is 0
6 Trace> QueryPlanOptimizePrewhere: Moved 1 conditions to PREWHERE
7 Debug> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Key condition: (column 0 in [777, 777])
8 Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Filtering marks by primary and secondary keys
9 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Running binary search on index range for part all_203_417_3 (29185 marks)
10 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (LEFT) boundary mark: 226
11 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (RIGHT) boundary mark: 227
12 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Running binary search on index range for part all_892_897_1 (816 marks)
13 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found continuous range in 29 steps
14 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Running binary search on index range for part all_886_891_1 (816 marks)
15 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (LEFT) boundary mark: 6
16 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Running binary search on index range for part all_898_903_1 (816 marks)
17 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Running binary search on index range for part all_634_849_3 (29321 marks)
18 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Running binary search on index range for part all_850_885_2 (4888 marks)
19 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Running binary search on index range for part all_1_202_3 (27419 marks)
20 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Running binary search on index range for part all_418_633_3 (29321 marks)
21 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (RIGHT) boundary mark: 7
22 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (LEFT) boundary mark: 6
23 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (LEFT) boundary mark: 227
24 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (LEFT) boundary mark: 298
25 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (RIGHT) boundary mark: 7
26 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (LEFT) boundary mark: 38
27 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (LEFT) boundary mark: 227
28 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Running binary search on index range for part all_904_909_1 (723 marks)
29 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found continuous range in 18 steps
30 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (LEFT) boundary mark: 5
31 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (RIGHT) boundary mark: 229
32 <Trace> hw15.transactions (e10e1615-0c33-4404-9974-cb87ba1938cd) (SelectExecutor): Found (RIGHT) boundary mark: 299

```

Логов по запросу, в котором поиск производится без индексов, на пять порядков больше:

```

30 [clickhouse-server-hw13] 2025.06.29 17:40:00.646331 [ 811 ] {09c7049f-a707-4e98-8288-647f5b175dea} <Trace> QueryConditionCache: Updated entry for table_id: e1
_1_202_3, condition_hash: 9972192576531345224, condition: , marks_count: 27419, has_final_mark: true
[clickhouse-server-hw13] 2025.06.29 17:40:00.647113 [ 811 ] {09c7049f-a707-4e98-8288-647f5b175dea} <Trace> MergeTreeSelectProcessor: Read steps statistics: st
09
[clickhouse-server-hw13] 2025.06.29 17:40:00.647671 [ 952 ] {09c7049f-a707-4e98-8288-647f5b175dea} <Trace> QueryConditionCache: Updated entry for table_id: e1
_1_202_3, condition_hash: 9972192576531345224, condition: , marks_count: 27419, has_final_mark: true
[clickhouse-server-hw13] 2025.06.29 17:40:00.650968 [ 967 ] {09c7049f-a707-4e98-8288-647f5b175dea} <Trace> QueryConditionCache: Updated entry for table_id: e1
_1_202_3, condition_hash: 9972192576531345224, condition: , marks_count: 27419, has_final_mark: true
[clickhouse-server-hw13] 2025.06.29 17:40:00.651301 [ 952 ] {09c7049f-a707-4e98-8288-647f5b175dea} <Trace> MergeTreeSelectProcessor: Read steps statistics: st
18
[clickhouse-server-hw13] 2025.06.29 17:40:00.651743 [ 967 ] {09c7049f-a707-4e98-8288-647f5b175dea} <Trace> MergeTreeSelectProcessor: Read steps statistics: st
18
[clickhouse-server-hw13] 2025.06.29 17:40:00.662817 [ 87 ] {09c7049f-a707-4e98-8288-647f5b175dea} <Debug> executeQuery: Read 1010000000 rows, 14.97 GiB in 20.4
iB/sec.
[clickhouse-server-hw13] 2025.06.29 17:40:00.708628 [ 87 ] {09c7049f-a707-4e98-8288-647f5b175dea} <Debug> MemoryTracker: Query peak memory usage: 130.50 MiB.
Showed 1000 out of 2768012 rows.

```